

Model for grain boundary sliding and its relevance to optimal structural superplasticity

Part 3 – Effect of flow localisation and specimen thickness on superplasticity in alloy Supral 100

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The effect of flow localisation and specimen thickness on the superplastic deformation of the commercial alloy Supral 100 (Al–6Cu) is discussed. Optimal flow was accompanied by the formation and spread of intense 'shear bands' along grain/interphase boundaries. The spacing between the shear bands depended on both the strain rate of deformation and the specimen thickness. Superplasticity drastically decreased when the specimen thickness fell below about 10 grain diameters. The apparent activation volume was a function of strain rate but was independent of specimen thickness, while the strain rate sensitivity index depended on both the strain rate and specimen thickness. The results are explained in terms of the theory that as well as being influenced by specimen geometry, the degree of superplasticity depends on the extent to which cooperative boundary sliding and boundary shear band formation are present during flow.

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Introduction

Supral 100 is a commercial superplastic alloy used extensively in sheet thermoforming. The superplastic properties of this alloy have been well documented.¹ In the thermoforming of sandwich structures the sheet may become extremely thin and give rise to failure in service. It is also known that the deformation behaviour of extremely thin sheets of superplastic alloys is different from that of thick sheets.² To the best of the present authors' knowledge, the deformation behaviour of very thin sheets of Supral 100 alloy has not been studied. It is of practical importance to understand the deformation behaviour since superplastic forming will be unsuccessful when the sheet thickness is below a certain value. Therefore, a study of the deformation behaviour of very thin sheets of this alloy was conducted to determine why the superplastic properties are altered when the sheet thickness is drastically decreased. A comparison was also made with the behaviour of thicker specimens. The difference in the deformation behaviour of the two types of specimens is accounted for.

Material and experimental procedures

Cold rolled sheets of Supral 100 (supplied by Alcan International, UK) were used. The sheets were 2–2.1 mm thick in the as received condition and had a composition (wt-%) of Al–5.99Cu–0.002Mg–0.06Si–0.16Fe–0.033Zn–0.004Ti–0.39Zr. Strips were cut from the sheets and rolled down further at room temperature using carbide rolls to thicknesses of 0.20, 0.10, 0.08, and 0.06 mm. Two microstructural conditions were developed by heat treatment. Annealing for 30 min at 450°C (723 K) gave an average two-dimensional grain size of $2.5 \pm 0.3 \mu\text{m}$, while annealing for 75 min yielded an average two-dimensional grain size of $3.5 \pm 0.3 \mu\text{m}$. (It must be noted that testing of superplastic materials in the laboratory at 450°C requires about 30 min for temperature equilibration, whereas industrial and technological processes usually take about 1 to 1.5 h for the stabilisation of temperature.) For the sake of comparison, a control specimen (which did not undergo any deformation) was also placed inside the furnace surrounding the

superplastic tensile testing system. The control specimen was removed from the furnace just before the start of deformation. Only the results from those specimens which had reproducible microstructures were used for further analysis.

Superplastic tensile tests were carried out on a Schenck Trebel electromechanical testing machine at a temperature of 450°C and an initial strain rate of $1.4 \times 10^{-3} \text{ s}^{-1}$ (constant crosshead speed tests) which lay in the region of optimal superplastic flow.¹ Tests for comparison were also carried out at strain rates of 1.4×10^{-2} and $1.4 \times 10^{-4} \text{ s}^{-1}$ in regions III and I of superplastic deformation, respectively. Microstructural studies were carried out using a Leitz Metallovert optical microscope. Metallographic specimens were prepared by polishing on a buffing wheel using diamond paste and subsequently etching with freshly prepared Keller's reagent. Unetched polished specimens were used to observe deformation relief. Fine scratches present on the surface of the specimens were used as marker lines to measure the shear deformation at boundaries. Strain rate jump tests were carried out at 25, 100, and 400% strain to determine the strain rate sensitivity index m . The apparent activation volume V^* was estimated using the well known relationship³

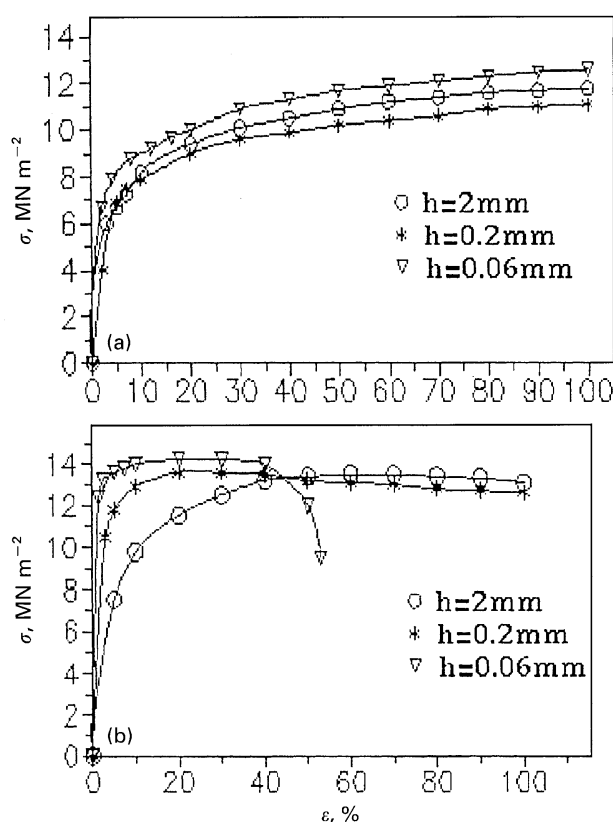
$$V^* = kT \left(\frac{\partial \ln \dot{\epsilon}}{\partial \sigma} \right)_{L,T} \dots \dots \dots (1)$$

where $\dot{\epsilon}$ is the true strain rate, σ is the true stress, L is the grain size, k is Boltzmann's constant, and T is the absolute temperature of deformation. The subscripts indicate the variables that were kept constant.

Results

MECHANICAL BEHAVIOUR

The mechanical behaviour of Supral 100, in particular that of material with a grain size of $2.5 \mu\text{m}$, was found to be similar to that found earlier. From the literature¹ it is seen that for a material of grain size $2.5 \mu\text{m}$, a strain rate of $1.4 \times 10^{-3} \text{ s}^{-1}$ falls in the range of optimal superplastic flow. In the present work it can be seen from the values of elongation to failure δ and strain rate sensitivity index m for both thick and thin specimens with a grain size of $3.5 \mu\text{m}$ (Table 1) that a strain rate of $1.4 \times 10^{-3} \text{ s}^{-1}$ does

a 2.5 μm grain size; b 3.5 μm grain size

1 Stress-strain plots for Supral 100 specimens of given thickness h for given grain size: test temperature 450°C, initial strain rate $1.4 \times 10^{-3} \text{ s}^{-1}$

fall within the optimal range of superplastic flow. Typical stress-strain plots are shown in Fig. 1. It could be seen from these plots that further study was necessary with regard to the initial portion of the stress-strain curve where the stress increases sharply with strain, and the steady state region of superplastic flow where the stress remains nearly constant with strain.

From Fig. 1 it was seen that the increase in stress with strain in the early part of the stress-strain curve was dependent on the thickness of the specimen being tested. With decreasing thickness, the stress rose more sharply with strain. This effect was clearly seen in specimens with a grain size of 3.5 μm (Fig. 1b). In specimens with a grain size of 2.5 μm (Fig. 1a) the effect was less significant: the stress-strain curves for the two thicker specimens practically coincided in the early part of the stress-strain curve (and in fact in the region beyond) and the flow curve for the specimen of intermediate thickness was lower than those for the other two thicknesses. A possible explanation for this is discussed below. Conversely, for the coarser grain size specimens, where a steady state was reached, i.e. the stress reached a constant value beyond which it did not

change with further strain, the strain needed to reach steady state flow increased with the specimen thickness. For specimens with smaller grain size, the stress increased continuously with strain (flow hardening arising mainly from microstructural evolution/grain coarsening). Figure 2 shows the variation of δ , m , and V^* , as functions of specimen thickness.

The value of δ decreased sharply when the thickness of the specimen was below 100 μm (Fig. 2a). For the alloy with a grain size of 3.5 μm , at a specimen thickness of 60–80 μm δ was less than 100% and the value of m was less than 0.3. These observations indicate that the superplastic behaviour had decreased drastically. For the same conditions, the change in V^* was negligible. The value of V^* was slightly lower for the alloy with a grain size of 3.5 μm compared with that for the alloy with a grain size of 2.5 μm (Fig. 2c). Although V^* appeared to be virtually independent of specimen thickness, the results given in Table 1 show that V^* was clearly dependent on the strain rate of deformation.

MICROSTRUCTURES

From an observation of the microstructure of the control specimen, it became clear that the Supral alloy had fine, equiaxed grains. The intermetallic particles were not present as chains but were distributed randomly. Before superplastic deformation the grain boundaries were not clearly visible even when freshly prepared Keller's reagent was used for etching. (Such a situation is typical of aluminium alloys which do not have a fully recrystallised microstructure.^{1,4,5}) In most cases the grain boundaries were more darkly etched after deformation. By careful observation of the microstructure and the etching characteristics after deformation, it was possible to determine the 'active' (i.e. where the deformation was concentrated) and the 'passive' grain boundaries, since the active boundaries were more darkly etched than the passive ones.

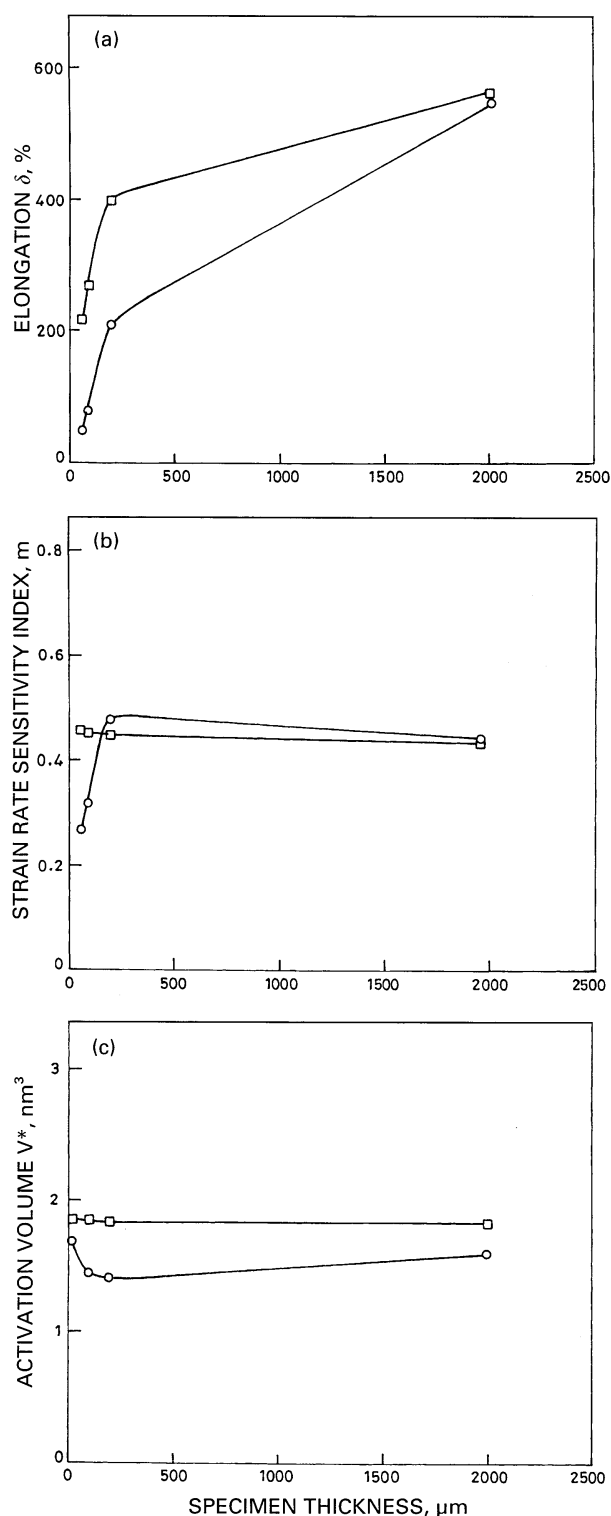
The overall microstructure of the specimen following deformation was as previously reported.¹ Grain elongation in the tensile direction, grain growth, and nucleation of cavities were seen in the microstructure. These results are given in Table 2. In addition, a few new observations were made in this study. It was seen that the active boundaries formed interlinked bands that were inclined to the tensile axis. In these bands, the boundaries had a very small angle of inclination between each other. The grain boundaries appeared to be aligned along the direction of these 'shear bands'. (In the literature these have also been described as 'deformation bands', but the term shear bands is to be preferred because in crystallographic terminology deformation band has a distinct meaning.⁶) Microcavities resulting from deformation were nucleated at these grain boundaries

Table 2 Average number of grains at failure within each domain bounded by interlinked microcavities in specimens of given thickness: test temperature 450°C, initial grain size 3.5 μm , initial strain rate $1.4 \times 10^{-3} \text{ s}^{-1}$

Specimen thickness, μm	Elongation at fracture δ , %	Average grain size at fracture d , μm	Average domain size H , μm	No. of grains per domain H/d
2100	540	7.7	79	10.2
2100	542	5.4	47	8.6
220	225	5.5	65	11.8
220	392	5.6	63	11.3
120	208	5.6	63	11.3
100	260	5.9	45	7.6
80	78	5.6	58	10.3
70	57	7.7	85	11.0
60	215	5.1	50	9.8
60	175	5.5	45	8.2

Table 1 Variation of elongation at fracture δ , strain rate sensitivity index m , and activation volume V^* for specimens of given thickness h (mm) deformed at given initial strain rates: test temperature 450°C, initial grain size 3.5 μm

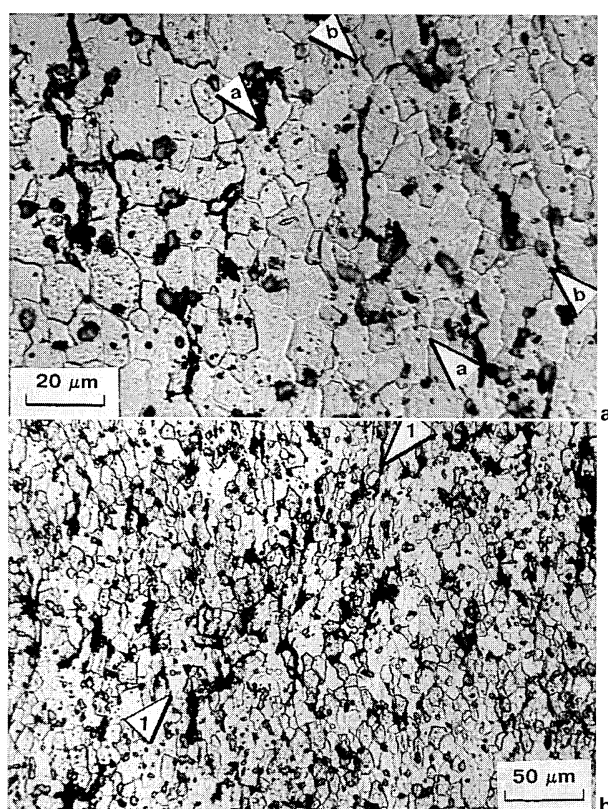
Initial strain rate, s^{-1}	δ , %		m		V^* , nm^3	
	$h=2$	$h=0.2$	$h=2$	$h=0.2$	$h=2$	$h=0.2$
1.4×10^{-2}	292	101	0.35	0.23	0.779	1.093
1.4×10^{-3}	540	225	0.43	0.49	1.602	1.405
1.4×10^{-4}	400	163	0.78	0.36	5.452	5.930



○ 3.5 μm grain size; □ 2.5 μm grain size

2 Variation of *a* percentage elongation to failure δ , *b* strain rate sensitivity index m , and *c* activation volume V^* , with specimen thickness in Supral 100: test temperature 450°C, initial strain rate $1.4 \times 10^{-3} \text{ s}^{-1}$

and appeared as bands or arrays of bands inclined to the tensile axis. This could easily be seen when the specimen was viewed under an optical microscope (Fig. 3). Figure 3a, which is at a higher magnification than Fig. 3b, reveals evidence for two cooperatively slid boundary bands that are shown by two mutually pointing arrows. Figure 3b shows a wide shear band containing a number of cavities.

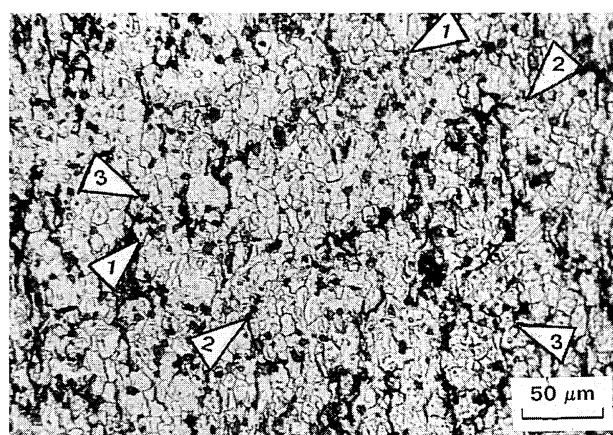


a cooperatively slid boundary bands inclined to tensile axis indicated by arrows; *b* shear band with cavitation indicated by arrows

3 Microstructures of Supral 100: average initial grain size 3.5 μm , test temperature 450°C, initial strain rate $1.4 \times 10^{-3} \text{ s}^{-1}$, specimen thickness 2.1 mm, tensile axis runs vertically up page

At high strain rates, the degree of cavitation was less and the cavities were also distributed more uniformly. At lower strain rates, cavitation was more pronounced and networks of cavity linked bands were clearly visible (Fig. 4). On large deformation the microcavities coalesced to form elongated macrocavities. These macrocavities were aligned along the tensile direction. Before fracture these macrocavities caused a significant reduction in the cross-section of the specimen and gave the specimen volume the appearance of being partitioned into different domains. The average cross-sectional size of these domains at the moment of fracture was 8–12 grain diameters (Table 2).

For the thin specimens the grain size was non-uniform after deformation. Bands of fine grains of about 6 μm with boundaries that etched dark were observed. These bands were separated by regions of grains with boundaries that etched poorly. In profile, well formed macronecks and shear bands were visible (Fig. 5). When the thickness of the specimen was reduced, cavitation also decreased. Fracture took place when the thickness of the specimen was about 10 grain diameters. Further insight was gained when the deformation relief of the prepolished specimen was studied after a strain of about 4–7%, i.e. in the region just before the onset of steady state superplastic flow. Thin wavy lines which covered the boundaries of several grains were visible on the surface of the specimen (Fig. 6). These lines seemed to be predominantly at 90° to the tensile axis. As explained in Part 2,⁷ with the aid of Fig. 3c of that paper, the lines are the trace on the specimen (length $\times$ width) plane of a boundary shear band inclined at 45° to the stress axis in the (length $\times$ thickness) plane. Horizontal as well as vertical displacements of the marker lines were observed on the wavy lines. Table 3 gives the frequency with which these bands were observed.



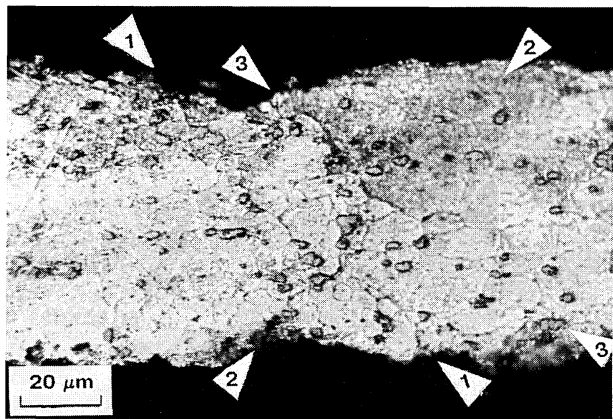
4 Microstructure showing cavity linked bands at the boundaries of dark etched grains indicated by arrows: average initial grain size $3.5\text{ }\mu\text{m}$, test temperature 450°C , initial strain rate $1.4 \times 10^{-4}\text{ s}^{-1}$, specimen thickness 2.1 mm , tensile axis vertical up page

Discussion

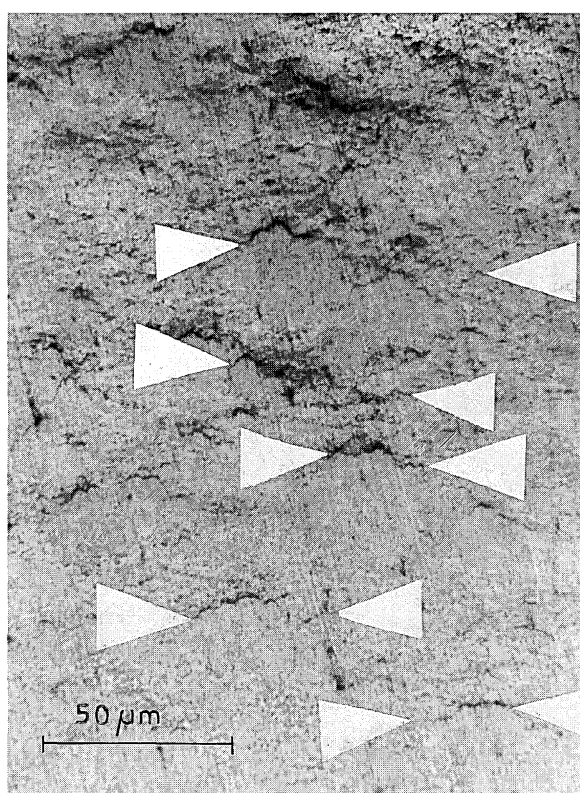
Deformation localisation during superplastic flow was found by one of the present authors in a model Zn–22 wt-%Al alloy.⁸ Therefore, it was decided to extend the study to a commercial superplastic alloy, namely, Supral 100. As noted in the introduction, for successful thermoforming of sandwich structures, the behaviour of very thin specimens has to be understood, and so the scope of the work was extended to include the effect of specimen thickness.

Supral 100 is processed to develop a fine grain microstructure by strain enhanced recrystallisation in the early stages of superplastic flow, as the pinning of subgrain boundaries by ultrafine ZrAl_3 precipitates confers on this material a very high resistance to static recrystallisation.⁹ The variable amount of cold deformation imparted to the specimens in the present study to obtain different thicknesses could have influenced the static and strain enhanced recrystallisation behaviour. Therefore the results have to be interpreted carefully.

A comparison of Fig. 1a and b reveals that true steady state flow conditions were established only when initial static recrystallisation had been sufficiently long to obtain a starting grain size of $3.5\text{ }\mu\text{m}$ (Fig. 1b). (In Fig. 1a there is continuous flow hardening, which is indicative of micro-



5 Well developed macronecks and shear bands, indicated by pairs of arrows, in profile: average initial grain size $3.5\text{ }\mu\text{m}$, test temperature 450°C , initial strain rate $= 1.4 \times 10^{-3}\text{ s}^{-1}$, specimen thickness 0.06 mm , tensile axis is horizontal



6 Deformation relief of prepolished specimen after 5.4% strain: wavy lines covering several grain boundaries that are indicative of development of cooperative boundary sliding are indicated by arrows: average initial grain size $3.5\text{ }\mu\text{m}$, test temperature 450°C , initial strain rate $1.4 \times 10^{-3}\text{ s}^{-1}$, specimen thickness 0.07 mm , tensile axis runs vertically up page

structural evolution.) Thus, the conclusions that in the early part of the stress–strain curve the stress rose more sharply with strain with decreasing specimen thickness, and that the strain needed to reach steady state flow increased with specimen thickness, are based on measurements corresponding to a single grain size ($3.5\text{ }\mu\text{m}$). The conclusions with regard to the effects of specimen thickness on δ , m , and V^* (see section on ‘Mechanical behaviour’ above), however, appear to be valid for both grain sizes, notwithstanding the microstructural evolution present in the material with starting grain size of $2.5\text{ }\mu\text{m}$.

It could be argued that in thinner tensile specimens the effects of inhomogeneities, e.g. thickness and compositional variations, would be relatively more significant, and that for a critical cross-section at failure a smaller reduction in area only would be necessary. This can account for the lower value of δ and the earlier peaking of the true stress as a function of true strain in the case of the thinner specimens. However, the higher flow stresses at a fixed strain observed for the thinner specimens (Fig. 1b) can not easily be explained by this line of argument, particularly in

Table 3 Average distance between boundary shear bands along longitudinal and transverse directions of tensile specimen: test temperature 450°C , initial grain size $3.5\text{ }\mu\text{m}$, specimen thickness 0.07 mm

Initial strain rate, s^{-1}	Average distance between shear bands, μm	
	Longitudinal direction	Transverse direction
0.014	9.8	12.5
0.0014	14.8	18.2
0.00014	17.4	23.5

view of the results in the literature that the strength of very thin material is greater than that of thicker sections. Further experimental work is necessary before it can be unequivocally established that the material properties are changed perceptibly when the specimen sections are very thin. For the present, it is suggested that this possibility can not be ruled out.

As the microstructural stability during superplastic flow was greater when the starting grain size was 3.5 μm than when it was 2.5 μm , further observations were made using only the 3.5 μm grain size (see Tables 1–3 and Figs. 3–6).

From the present experimental data as well as the previously published results¹ the formation and spread of intense shear bands at grain boundaries during optimal superplastic flow has been clearly demonstrated.

In the initial stage of deformation, plane interfaces that connected two or more grain/interphase boundaries during optimal superplastic flow, and bands that were the result of cooperative boundary sliding, formed at different places in the tensile specimen surface as well as in the interior of the specimen. Over a significant distance, such a band appeared as a wavy interface inclined to the loading direction. (A number of plane interfaces of shorter length joined together and became smoothed during further sliding to give the wavy appearance of the shear bands (see Part 2).⁷) This interface comprised parts of grain boundaries that were oriented favourably to the maximum shear stress direction. This was evident from the deformation relief studies of the surface of the specimen and the microstructural analysis in the bulk of the deformed specimen, in particular from the difference in the etching characteristics of different grain boundaries. It is known that the degree of etching of grain boundaries is dependent on the energy of the boundaries. When the energy is greater, the grain boundaries are etched more darkly. Boundaries with high energy are more mobile in general and more suitable for grain/interphase boundary sliding compared with boundaries of lower energy, e.g. small angle boundaries and special boundaries. It is known that the energy of a boundary can increase as a result of grain/interphase boundary sliding. Thus, it may be concluded that the dark etching (high energy) interconnected boundaries were the result of cooperative boundary sliding. These boundaries tended to become aligned more or less linearly along a shear band and facilitated continual cooperative boundary sliding (see Part 2, Ref. 7). It is at these favourably oriented boundaries that sliding started. An increase in the applied stress would be necessary to overcome the local stress concentrations created at unfavourably oriented boundaries in order to facilitate the continuation of cooperative boundary sliding along those boundaries.

When the volume of the specimen was large, the probability of the development of favourably configured shear bands for continual boundary sliding was high, and therefore the rate of increase in stress with deformation in the initial stage of deformation was low. When the thickness of the specimen was about 10–20 grain diameters, the probability of generating a favourable configuration for cooperative boundary sliding was reduced and so the rate of increase of stress with deformation was higher. This makes sense because a higher stress would be necessary for cooperative boundary sliding to take place along the reduced number of bands available in the specimen. Using this argument it becomes possible to explain qualitatively the differences in the initial part of the stress–strain curves obtained for different specimen thicknesses (Fig. 1b). However, it must be emphasised that this argument only holds good for the case of cooperative boundary sliding, as a minimum of about 10 grains initially in the through thickness direction of the specimen appeared to be necessary for cooperative sliding to take place in a significant manner. (It is possible to argue that when the specimen is less than

10 grains thick the shear bands need not be as long as in thicker specimens, but the results, which are presented in Table 2, do not provide support for this argument. Thus, at least in the range of thicknesses presently studied, the minimum length of the shear bands formed seems to be independent of specimen thickness.)

When the specimen thickness dropped to less than about 10 grain diameters, superplasticity was drastically decreased and the fracture of the specimen was similar to that seen during conventional high temperature fracture. (Scanning electron microscopy studies of this will be reported elsewhere.) This conclusion could also be drawn from the microstructure of the fractured sample in which the specimen thickness was less than 10 grain diameters (see, for example, Fig. 10 of Paton and Hamilton¹⁰). This observation explains why the elongation decreased when the specimen became thin. In the case of thick specimens, the overall thickness was less important than the cross-section of the specimen ligaments, i.e. the area between large interlinked cavities. Similarly, the distance between the voids or holes, as in the experiments of Tait and Taplin,¹¹ and between the reinforced fibres in the case of composites, became extremely important compared with the bulk thickness of the specimen.

When the number of shear bands was large enough for continual cooperative boundary sliding to be achieved, the steady state flow region of superplasticity sets in. In this condition the shear bands were transformed into macrobands along which intense deformation took place (the boundary sliding equivalent of Chernov–Lüder's bands). The width of these bands was of the order of a few grain diameters. In these bands, apart from boundary sliding, rotation of grains and cavitation were also seen. From the shape of the cavities and their linkages, it was clear that extensive deformation also took place along these bands in the interior of the specimen. Between these bands were zones that underwent only a relatively small degree of deformation. The number of the bands and the size of the passive zones depended on the strain rate of deformation and the thickness of the specimen. When the strain rate or the thickness of the specimen was reduced, there was a greater degree of non-uniform deformation. (With a decrease in strain rate (i.e. stress), the number of grain boundaries at which the critically resolved shear stress necessary to bring about cooperative boundary sliding was available, decreased.) Further work is necessary to include the flow localisation reported here quantitatively into the mechanistic model presented in Part 1.¹²

It is seen from Table 1 that V^* is a function of the applied strain rate and that it increases with decreasing strain rate. If superplasticity is explained in terms of stimulated boundary sliding and the movement of grain boundary dislocations, as has been done by Kaibyshev and co-workers,^{9,13} then it could be suggested that, at lower strain rates (or stresses), fewer lattice dislocations will originate at a boundary and traverse the grain to become absorbed at an opposite boundary. Thus, the grain boundary structure is disturbed to a lesser extent and V^* is larger. At higher strain rates more lattice dislocations will run into the opposite boundaries and stimulate sliding. This process will reduce the size of V^* . This approach has yet to be quantified.

In contrast, in the approach of Padmanabhan and Schlipf,¹² from their equations (12) and (30) the strain rate during optimal superplastic flow $\dot{\epsilon}_{sp}$ is given by

$$\dot{\epsilon}_{sp} = 2\nu \exp\left(\frac{-\Delta F_0}{kT}\right) \frac{W}{L} \frac{e_0^2 V_0}{kT} \times \left\{ (\sigma - \sigma_0) \left[1 + \alpha \left(\ln \frac{\sigma - \sigma_0}{\sigma_m} - 1 \right) \right] + \sigma'_1 \alpha \right\} \quad (2)$$

where $\alpha = 2/\lambda\pi^{1/2}$ and $\sigma'_1 = \sigma_m \exp(-\lambda\pi^{1/2}/2)$, and σ_m and

λ are the mean and standard deviation, respectively, of the internal stress distribution resulting from mesoscopic cooperative sliding. (For other definitions see Ref. 12.) As equation (1) defines V^* as equal to $kT(\partial \ln \dot{\epsilon}/\partial \sigma)$, it follows that in this analysis

$$V^* = kT \left(\frac{1 + \alpha \ln \frac{\sigma - \sigma_0}{\sigma_m}}{(\sigma - \sigma_0) \left[1 + \alpha \left(\ln \frac{\sigma - \sigma_0}{\sigma_m} - 1 \right) \right] + \sigma'_1 \alpha} \right) \quad (3)$$

Since the strain rate is a function of σ , it is natural that V^* depends on both $\dot{\epsilon}$ and T . The activation volume will reflect the properties of the basic unit of sliding (an oblate spheroid of volume V_0 defined in Ref. 12) via λ and σ_m , but it should not be identified with V_0 . The latter would be obtainable experimentally only from a more sophisticated measurement or theoretically from atomistic calculations based on molecular dynamics. (The dependence of V_0 on $\dot{\epsilon}$ and T is yet to be determined.)

It can also be seen from Fig. 2c that V^* remains relatively unchanged when the specimen thickness is changed, whereas m is dependent on specimen thickness. This is explained by the fact that V^* concerns the micromechanism of boundary sliding, which is independent of the thickness of the specimen being tested. However, m is indicative of the degree of cooperative boundary sliding, i.e. plane interface formation (see Parts 1 and 2, Refs. 12 and 7), and the formation of networks of shear bands to ensure smooth and continual sliding over large areas. Since the formation of these bands is dependent on the thickness of the specimen (see above), it can easily be seen that m would be strongly dependent on the thickness of the specimen, which is what has been observed. It is noted at this stage that the propensity for neck formation changes with specimen thickness. Therefore, this geometric factor will also have to be considered when deriving expressions for the effect of specimen thickness on the value of m . (In the model of Padmanabhan and Schlipf, m can be obtained from equation (29) of Ref. 12 ($m = n^{-1}$). The effect of specimen thickness on the value of m is yet to be included in this analysis.) Therefore, a major conclusion of this study is that V^* is a measure of the micromechanism of boundary sliding, whereas m is a measure of the extent of cooperative sliding present in the deforming specimen. This explains why in the case of high temperature power law creep of alloys, although boundary sliding is present and V^* is also nearly of the same order as that for superplastic flow, the value of m is rather low. Evidently, this is due to the degree of cooperative sliding being considerably less during high temperature power law creep than during optimal superplastic flow. For coherent deformation in the former case, in addition to boundary sliding less strain rate sensitive dislocation processes will also be necessary and this will lower the overall value of m .

Conclusions

1. Intense shear bands formed and spread along grain/interphase boundaries during optimal superplastic flow. This is attributed to cooperative boundary sliding involving a number of grain/interphase boundaries. This process, which connects the differently oriented boundaries by sliding to form a continuous network in which deformation becomes concentrated, is responsible for the strong dependence of stress on strain in the early stages of superplastic flow.

2. The spacing between the boundary shear bands was a function of both the strain rate of deformation and the

thickness of the specimen. When the specimen thickness was less than about 10 grain diameters, cooperative sliding and boundary shear band formation became difficult and this led to a significant decrease in superplasticity.

3. The apparent activation volume is independent of specimen thickness as it is an indicator of the micromechanism of boundary sliding which does not depend on specimen thickness. However, the activation volume is dependent on the strain rate of the deformation and this can be explained in two different ways.

4. The strain rate sensitivity index m depended on the specimen thickness as well as the strain rate as it is a measure of the degree to which cooperative boundary sliding is present during superplastic flow. This explains why m is relatively low in high temperature power law creep deformation even though boundary sliding is present. This is because in creep cooperative boundary sliding/boundary shear band formation is not significant and so coherent strain can only be produced with the aid of an additional low strain rate sensitive mechanism, e.g. dislocation motion. This leads to a lowering of the overall m value in high temperature creep.

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